



DEALER BEST PRACTICE SERIES

CRC Planning and
Scheduling Systems

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

Efficient planning and scheduling of CRC manpower and resources to the workload, and adjusting for constant changes is a major challenge for most dealerships. Caterpillar has no preferred software available to dealers to perform this function. Often, planning and scheduling of work orders is performed on a display board or on an Excel worksheet. Supervisors spend much of their day managing work assignments and workload in their areas.

When a CRC has more than 50 technicians, and especially when there is more than one shift, a formal scheduling process and supporting software becomes a necessity.

Several dealers have found and implemented software designed to facilitate planning and scheduling. These software programs come from systems used in manufacturing plants and from the airline industries to manage many resources and tasks in a constantly changing environment.

This Best Practice discusses a scheduling process implemented at Finning Chile.

2.0 Best Practice Description

The dealer objective in implementing a new planning and scheduling process was to improve customer satisfaction with component rebuilds completed in a time, predictable manner. At the same time, to increase labor efficiency and CRC resource utilization by improving short and medium term planning and scheduling processes in the CRC, Machine Shop Salvage, and Parts deliveries.

The pain:

- Lack of coordination between the CRC and machine shop support.
- Lack of the ability to track and monitor work order status and completion target.
- Poor use of available resources – bottlenecks and delays.
- Unknown actual current and future work load
- Ineffective communication between planners, foremen, technicians
- Work order completion forecast seldom achieved.
- Lack of scheduling and coordination of subcomponent and the major component rebuild.
- Lack of coordination between the arrival of a component and the start of the rebuild.

The major steps in implementing a new scheduling process and system include:

- Create a team to work on the project.
- Study software packages available and select supplier.
- Create detailed process maps for most common rebuild activities. Each separate activity that requires a resource must be identified. Standard times for each segment are necessary.
- Define each resource. This includes technicians and workbays, but can also include major tooling, such as test bench time, dyno time, overhead crane, etc.

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- Define inter-dependence of each activity. This means tasks that cannot be started until a previous task is complete, and those tasks that can be performed at the same time. (Gantt Chart)
- Install and customize software.
- People training as required.

3.0 Implementation Steps

There are several dealers who are using scheduling software from suppliers. There are currently suppliers with systems operating very successfully. Including (but not limited to):

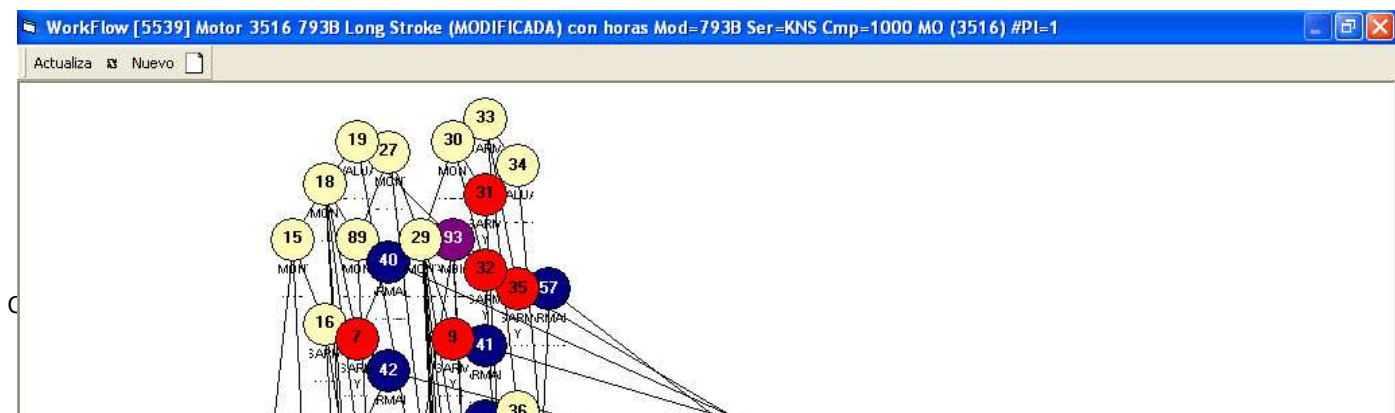
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- Ultimus
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Caterpillar does not endorse these particular suppliers; simply some dealers have successful experience working with them. Caterpillar welcomes additional suggestions from our dealer network.

The most difficult part of the implementation of a scheduling system is to define each rebuild activity (segments) in a process map. The activity must define the task and the time required. The activity needs to include parts ordering and delivery, and work performed in the machine shop.

Many process maps will be similar, such as for 3500 series engines, so that a standardize template can be used. Some models may have to be customized.

Sample 3516 Process Map:



Receiving /
Cleaning /
Disassembly

Assembly / Paint/
Testing / Shipping

Subcomponents
Machine Shop
Parts

Another important step is to create a Gantt chart of the entire process, which defines which segments must wait for a previous segment to complete, and which segments can be performed at the same time in different bays / locations, and how much time each segment requires.

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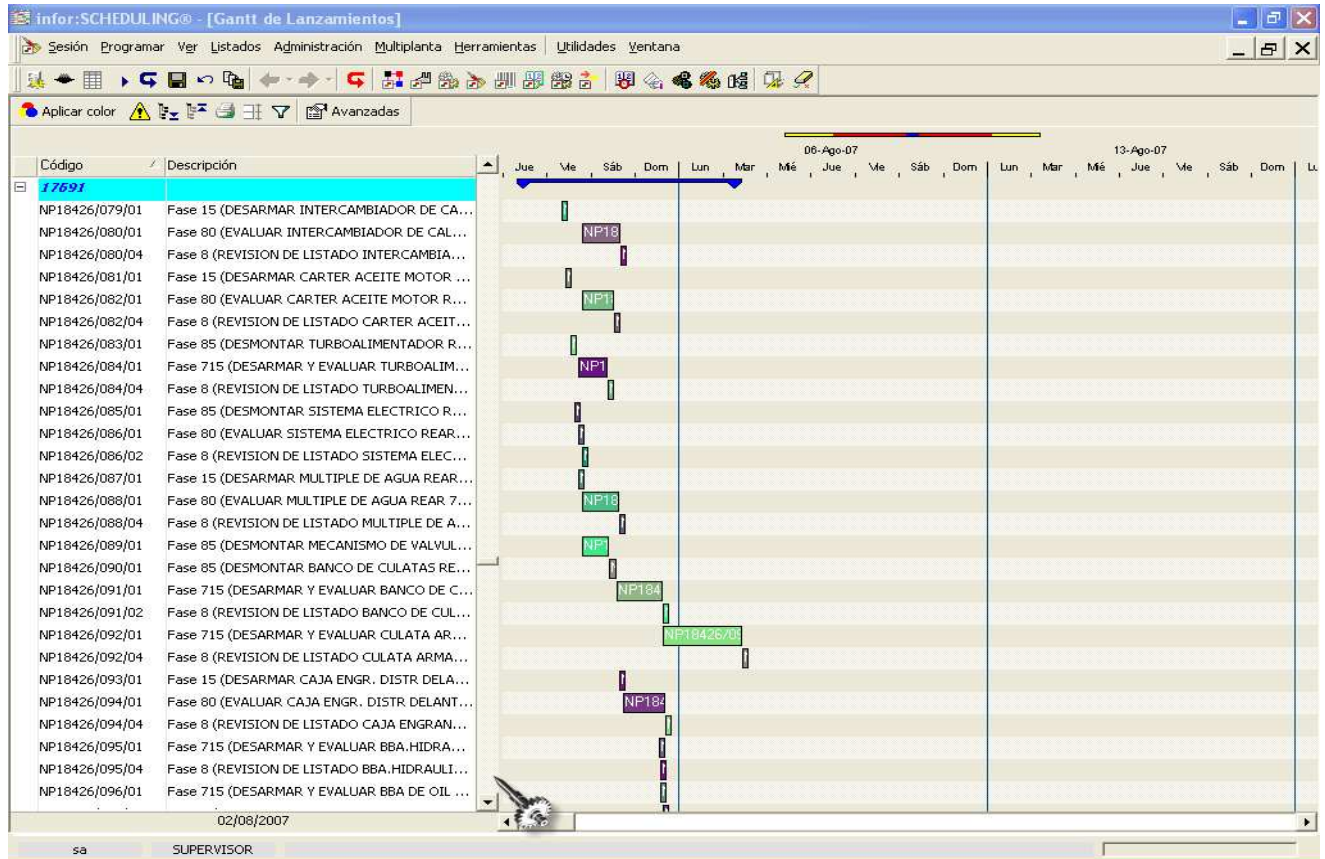
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Sample Gantt Chart



When the process mapping process is complete, the scheduling software can be installed and program to match each task to a resource and produce a schedule.

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Sample Technician Schedule

FINNING CAT		3-PT Transmisiones		02/08/2007			
WO	Operación	Estación	HH Prog	Inicio	Fin	OK HH Faltan	Observaciones
NP18327 010/01	16 - ARMAR TRANSMISION 777D	3-Transmisiones Recurso 1	1,00	02/08/07 08:00	02/08/07 09:00		
NP18346 010/01	16 - ARMAR TRANSMISION 793C	90-Transmisiones Recurso 2	20,00	02/08/07 08:00	03/08/07 18:00		
NP18414 005/01	16 - ARMAR PLANETARIOS TRANSMISION 797B	91-Transmisiones Recurso 3	9,00	02/08/07 08:00	02/08/07 18:00		
NP18428 010/01	16 - ARMAR TRANSMISION 793B	92-Transmisiones Recurso 4	4,00	02/08/07 08:00	02/08/07 12:45		
NP18448 010/01	16 - ARMAR TRANSMISION 793B	3-Transmisiones Recurso 1	25,97	02/08/07 09:01	04/08/07 13:44		
NP18428 013/01	13 - INSTALAR GRUPO DE CTROL EN TRANSMIS	92-Transmisiones Recurso 4	1,00	02/08/07 12:46	02/08/07 13:46		
NP18557 002/01	715 - DESARMAR Y EVALUAR FRENO PARQUEO 83	92-Transmisiones Recurso 4	8,00	02/08/07 13:47	03/08/07 10:02		
NP18414 007/01	16 - ARMAR TRANSMISION 797B	91-Transmisiones Recurso 3	23,98	02/08/07 18:01	05/08/07 08:00		

Work Order Number

Activity

Work Station

Required Time
Start time
Finish Time

"Complete"
"Not Initiated" or
"Need More Time"

02/08/2007
 16:38:32
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Informe Tarea
 [X] Sin Comenzar
 [tilde] Cumplida
 [Horas] Trabajo Inconcluso - Indicar Horas Faltantes

Turno Código Operador Nombre Firma

Near the end of each shift, the technician turns in his schedule for the day. Each task is marked either as "Done", "Not Initiated", or "Need X amount more time".

The information is entered into the schedule program, and a new schedule created based on progress and any changes.

Unplanned changes, such as an unplanned technician absence, can be entered into the schedule as it occurs and work for that day will be automatically redistributed.

4.0 Benefits

- All available manpower is fully scheduled and utilized.
- Schedule can show shop workloads months into the future and help plan incoming workloads.
- Management reports show area manpower utilization and bottlenecks. Enables corrective measures to be taken.
- Component turnaround time in the CRC is greatly reduced. Capacity is improved.
- Supervisors do less crisis management.
- Customers can have an exact status check on their work and expected completion time in seconds.

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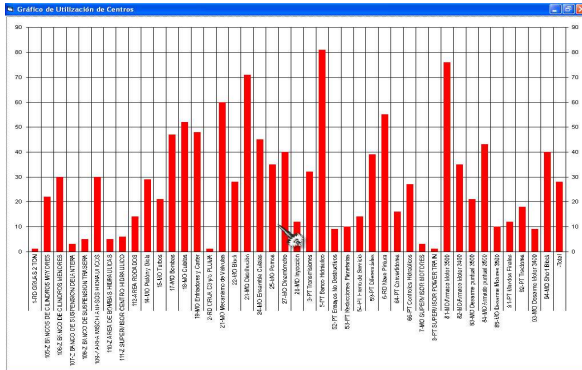
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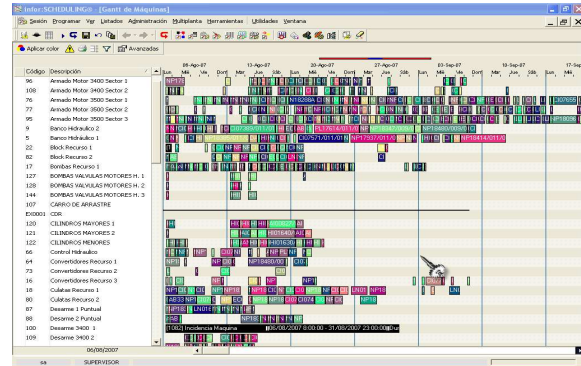
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- Improved communications – all departments.
- Technicians know daily schedule and standard times for each job.
- Allows more regular tooling maintenance and manages interruptions due to maintenance.



Utilization Report



Bottleneck Report

5.0 Resources Required

- Software cost with installation.
- Team to develop process map templates for common repairs.
- Total time- start to finish one year.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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